Theorem Book - Probability

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1 Introduction

This is a summary of the content in the first half of MATH10013, note that $\mathbb{N} = \{1, 2, 3...\}$, $\exp(x)$ is e^x.

2 Probability Axioms and Immediate Results

Axiom 2.1. A **probability space** is equipped with a set Ω, called the **sample space** and a **probability measure**, $\mathbb{P}$, which is a function with Subsets of $\Omega \to \mathbb{R}$. Such subsets are called **events**, and $\mathbb{P}$ have the following properties:

- $\mathbb{P}(\Omega) = 1$
- For any events A, $0 \leqslant \mathbb{P}(A) \leqslant 1$
- For countably finite events $A_1, A_2, A_3...$, if they are pairwise disjoint (i.e. $A_i \cap A_j = \emptyset$ for all $i \neq j$), then we have

$$\mathbb{P}(A_1 \cup A_2 \cup A_3...) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(A_k) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \mathbb{P}(A_3)...$$

Theorem 2.2. Some immediately provable results for any events $A, B$are as follows:

- $\mathbb{P}(\emptyset) = 0$
- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

Theorem 2.3 (Inclusion-Exclusion Principle)**.** For any events $A_1, A_2, A_3, ...$:

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \left[(-1)^{k+1} \times \sum_{S:|S|=k} \mathbb{P}\left(\bigcap_{j \in S} A_j\right)\right]$$

While the rightmost sum means 'choose k number and sum all the probability of intersections of such k set'. Theorem 2.2 is a example of the principle while $n = 2$.

Theorem 2.4 (de Morgan's Law)**.** For sets $A_1, A_2, A_3...$, we have

$$\left(\bigcup_{k=1}^{\infty} A_k\right)^c = \left(\bigcap_{k=1}^{\infty} A_k^c\right) \qquad \text{and} \qquad \left(\bigcap_{k=1}^{\infty} A_k\right)^c = \left(\bigcup_{k=1}^{\infty} A_k^c\right)$$

For example, in the case of $n = 2$, we have $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

3 Basic Combinatorics and Probability, Dependence of Events

Theorem 3.1 (Equally likely outcomes)**.** If the sample space Ω is finite, and given every element ω_k for $1 \leqslant k \leqslant n$ we have $\mathbb{P}(\omega_k)$ equal, then given a set $A \in \Omega$, we have

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}$$

Definition 3.2. When permuting r objects from n choices, we denote the number of available permutation to be P_r^n, with the formula of $P_r^n = \dfrac{n!}{(n-r)!}$

Definition 3.3. When choosing r objects from n choices (such that order doesn't matter), we denote the number of available combination to be $\binom{n}{r}$, with the formula of $\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$

Definition 3.4. The multinomial coefficient is

$$\binom{n}{k_1, k_2, k_3 ... k_r} = \frac{n!}{k_1! k_2! k_3! ... k_r!}$$

With the restriction $k_1 + k_2 + ... k_r = n$. Note the binomial coefficient is the multinomial coefficient $\binom{n}{r,n-r}$.

Theorem 3.5 (Binomial Theorem)**.**

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$$

Substitute $a = 1, b = 1$ we can easily see that $2^n = \sum_{k=0}^{n} \binom{n}{k}$.

Theorem 3.6 (Multinomial Expansion)**.**

$$(a_1 + a_2 + ... + a_k)^n = \sum_{i_1+i_2+...i_k=n} \binom{n}{i_1, i_2 ... i_k} a_1^{i_1} a_2^{i_2} ... a_k^{i_k}$$

Where the sum $\sum_{i_1+i_2+...i_k=n}$ generates overs all combination of $(i_1, i_2 ... i_k)$.

Definition 3.7. For two events A, B, if $\mathbb{P}(B) > 0$, then we define **the probability of A given B** to be

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Theorem 3.8. For an events B such that $\mathbb{P}(B) > 0, \mathbb{P}(\cdot|B)$ is a probability space. That is, For every theorem we developed on $\mathbb{P}(X), \mathbb{P}(Y)...$, it should works on $\mathbb{P}(X|B), \mathbb{P}(Y|B)...$ too. Note that this includes discussion on **expectation** too.

Theorem 3.9 (Partition Theorem)**.** If A_i is a disjoint union of the sample space Ω, then for event B,

$$\sum_{i=1}^{n} \mathbb{P}(B|A_i)\mathbb{P}(A_i) = \mathbb{P}(B)$$

This is explaining how the 'tree diagram' method works.

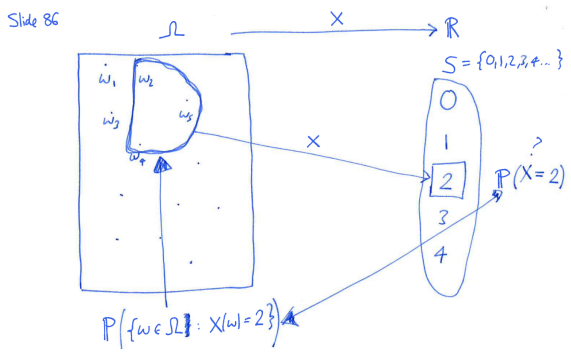
Definition 3.10. Event $A_1, A_2 ... A_n$ are **independent** if

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} \mathbb{P}(A_i)$$

In particular, $\mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A \cap B) \Leftrightarrow A, B$ are independent.

4 Single Random Variables

Definition 4.1. A **random variable** X is a function $\Omega \to \mathbb{R}$, if the output of X is countable, say $X(\Omega) = S = \{x_1, x_2, ...\}$, then X is a **discrete random variable**.



The intuition of a discrete random variable can be seen in the above figure.

Definition 4.2. A **probability mass function (pmf)** $p_X(x)$ for discrete random variable X, is defined as

$$p_X(x) = \mathbb{P}(X = x)$$

To check a valid pmf check $\sum_x p_X(x) = 1$. Note that the $\sum_x$ symbol remind you to sum any values of x that have non-zero probability.

Definition 4.3. If S is uncountable, or if there exists $f_X(x)$ for a random variable X, such that

$$\int_a^b f_X(x)\,dx = \mathbb{P}(a < X < b)$$

then X is a **continuous random variable** and $f_X(x)$ is the respective **probability density function**.

To check a valid pmf, check $f_X(x) \geqslant 0$ for all x and $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$. Again, the upper and lower bound of integration can be reduced to their non-zero part.

Definition 4.4. The **expected value / mean** of a random variable X, denoted by $\mathbb{E}(X)$, is

$$\mathbb{E}(X) = \begin{cases} \sum_{x \in S} x p_X(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x f_X(x)\,dx & \text{if } X \text{ is continuous} \end{cases}$$

Moreover, if $Z = g(X)$ for $g : \mathbb{R} \to \mathbb{R}$, then

$$\mathbb{E}(Z) = \begin{cases} \sum_{x \in S} g(x) p_X(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x) f_X(x)\,dx & \text{if } X \text{ is continuous} \end{cases}$$

The **variance** of X, often denoted by $\mathrm{Var}(X)$ or σ^2 if no ambiguity arises, is

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$$

Definition 4.5. The **probability distribution function / cumulative distribution function (cdf)** of a random variable X is

$$F_X(x) = \mathbb{P}(X \leqslant x) = \int_{-\infty}^{x} f_X(t)\,dt \qquad \text{if } X \text{ is continuous}$$

Theorem 4.6 (Transforming Continuous RV)**.** Given continuous random variable X, if $Z = g(X)$, where $g : \mathbb{R} \to \mathbb{R}$, then Z is a random variable, and the density function

$$f_Z(z) \begin{cases} f_X\left[g^{-1}(y)\right] \times \left[\frac{d}{dy}\left(g^{-1}(y)\right)\right] & \text{when } g \text{ is strictly increasing} \\ -f_X\left[g^{-1}(y)\right] \times \left[\frac{d}{dy}\left(g^{-1}(y)\right)\right] & \text{when } g \text{ is strictly decreasing} \end{cases}$$

The better way to think about this theorem is to write out the cdf

$$F_Z(z) = \mathbb{P}(Z \leqslant z) = \mathbb{P}[X \leqslant g^{-1}(z)]$$

then find the density.

Famous distributions, mean and variance for random variables X are listed below.

Name of X	Parameters	Values of x	PMF / Density	$\mathbb{E}(X)$	$\text{Var}(X)$
Bernoulli	Bernoulli(p)	$x = 0, 1$	$p_X(x) = \begin{cases} p & \text{when } x = 1 \\ 1-p & \text{when } x = 0 \end{cases}$	p	$p(1-p)$
Binomial	Bio(n, p)	$x = 0, 1, 2, ...n$	$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Geometric	Geom(p)	$x = 1, 2, 3...$	$p_X(x) = (1-p)^{x-1} p$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$
Poisson	Poi(λ)	$x = 0, 1, 2, 3...$	$p_X(x) = \lambda \frac{e^{-\lambda x}}{x!}$	λ	λ
Uniform	$U(\alpha, \beta)$	$x \in [\alpha, \beta]$	$f_X(x) = \frac{1}{\beta - \alpha}$	$\frac{\alpha+\beta}{2}$	$\frac{1}{12}(\beta-\alpha)^2$
Exponential	Exp(λ)	$x > 0$	$f_X(x) = \lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma	Gamma(α, λ)	$x > 0$	$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Normal	$\mathcal{N}(\mu, \sigma^2)$	$x \in \mathbb{R}$	$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$	μ	σ^2

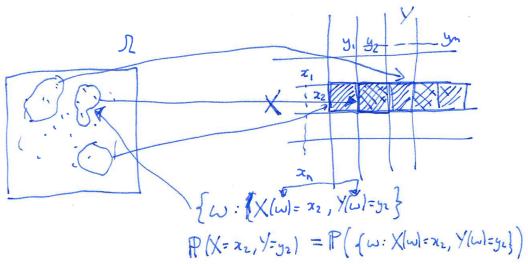
Notes of different distribution:

- Bernoulli is the basic object, it indicates the success or failure of a single event with probability p. (Example: Flipping a coin)
- The binomial distribution is when n independent Bernoulli trials was carried out, then X will be the number of success. (Example: Flipping n coins)
- Geometric distribution is when a Bernoulli trial is repeatedly carried out until success (then immediately stops), X will be the number of experiments done for this to happen. (Example: Flipping n coins until a head comes up)
- Poisson distribution is modeling how a rare event will occur given a time interval, with "expected" occurring rate of λ. (e.g. 600000 cards given a week, winning probability is $\frac{1}{200000}$, the number of winning tickets given week can be modeled by Poi(3). - check problem sheet)
- The exponential distribution is a **continuous analogue** of the geometric distribution. They can even be defined by the following property:
  - **Memoryless Property.** A random variable X is **memoryless** if given any number $s, t \in \mathbb{R}$:
    $$\mathbb{P}(X > s + t | X > t) = \mathbb{P}(X > s)$$
    Intuitively, it means "previous failures for any time doesn't affect future success".
- The Gamma function $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx$, if $\alpha \in \mathbb{N}$ then $\Gamma(\alpha) = (\alpha - 1)!$ (factorial)
- Gamma$(1, \lambda)$ is Exp(λ).

5 More than one Random Variables

Definition 5.1. A **joint probability mass function** consisted of two discrete random variables X, Y, where

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$$



Again, a intuition can be shown above, to check for a joint pmf and prove a lot of things related to it: we check $\sum_x \sum_y p_{X,Y}(x, y) = 1$, note that the double sum can be switched to prove important results. Similarly,

Definition 5.2. A **joint probability density function** consisted of two continuous random variables X, Y, where

$$\int_c^d \int_a^b f_{X,Y}(x, y)\, dx\, dy = \mathbb{P}(a < X < b, c < Y < d)$$

Take note on the integration order and relation to the probability. As with joint pmf, summing non zero probability sometimes may mean doing an area integral.

Theorem 5.3. The **marginal** of a random variable X, is the pmf/pdf of X, which can be recovered by

$$p_X(x) = \begin{cases} \sum_y p_{X,Y}(x, y) & \text{if } X \text{ continuous} \\ \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy & \text{if } X \text{ continuous} \end{cases}$$

Definition 5.4. For random variables X, Y, the **conditional probability mass/density function** of X given Y is:

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} \qquad \text{or} \qquad f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

Note that this is a valid mass/density.

One can now see linkage between notion of random variables and events. Independence are discussed similarly.

Definition 5.5. Two random variables are **independent** if $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ for all $x, y \in \mathbb{R}$.

Similar definition for more than 2 variables.

Definition 5.6. Random variables $X_1, X_2 ... X_n$ are **independent and identically distributed (iid)** if we have $p_{X_i}(x) = p_{X_j}(x)$ for all $1 \leqslant i, j, \leqslant n$, and

$$\prod_{i=1}^{n} p_{X_i}(x_i) = p_{X_1, X_2, ... X_n}(x_1, x_2, ... x_n)$$

Theorem 5.7 (Convolution)**.** If X, Y are independent random variables, then their sum $X + Y$ can be found by

$$p_{X+Y}(z) = \begin{cases} \sum_{i=-\infty}^{\infty} p_X(z - i)p_Y(i) & \text{if } X, Y \text{ discrete} \\ \int_{-\infty}^{\infty} f_X(z - y)f_Y(y)\, dy & \text{if } X, Y \text{ continuous} \end{cases}$$

6 Expectations, Variance and Covariance

Definition 6.1. The **covariance** of two random variable X, Y are:

$$\text{Cov}(X,Y) = \mathbb{E}\left[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))\right] = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

Where the expectation $\mathbb{E}(XY)$ can be calculated as

$$\sum_x \sum_y xy p_{x,y}(x,y)$$

given other random variables with $g(X,Y), \mathbb{E}(g(X,Y))$ can be calculated similarly.

Definition 6.2. The **correlation coefficient** $\rho(X,Y) = \dfrac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$, note that ρ is normalized $(-1 \leqslant \rho \leqslant 1)$ so it can be compared across different X, Y.

Theorem 6.3. For every random variable X, Y, and $a, b \in \mathbb{R}$:

- $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$
- $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$
- $\mathbb{E}(X)\mathbb{E}(Y) = \mathbb{E}(X)\mathbb{E}(Y) \Rightarrow X, Y$ are independent
- $\text{Var}(aX + b) = a^2\text{Var}(X)$
- $\text{Var}(X + Y) = \text{Var}(X) + 2\text{Cov}(X,Y) + \text{Var}(Y)$
- $\text{Cov}(X,X) = \text{Var}(X)$
- $\text{Cov}(Y,X) = \text{Cov}(X,Y)$
- $\text{Cov}(X + c, Y) = \text{Cov}(X,Y)$ for all $c \in \mathbb{R}$
- (Chebyshev's inequality) For any constant $c \in \mathbb{R}$, $\mathbb{P}(|X - \mu| < c) \leqslant \dfrac{\text{Var}(X)}{c^2}$, where $\mu = \mathbb{E}(X)$.

Moreover, if X, Y are independent,

- $\mathbb{E}(X)\mathbb{E}(Y) = \mathbb{E}(XY)$
- $\mathbb{E}(g(X)h(Y)) = \mathbb{E}(g(X))\mathbb{E}(h(Y))$ for any functions $g, h : \mathbb{R} \to \mathbb{R}$
- $\text{Cov}(X,Y) = 0$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Note that $\text{Cov}(X,Y) = 0$ or $\mathbb{E}(X)\mathbb{E}(Y) = \mathbb{E}(XY)$, does not implies the independence of X, Y , always check $p_X(x)p_Y(y) = p_{X,Y}(x,y)$.

Theorem 6.4 (Weak Law of Large Number)**.** Note that if $X_1, X_2 ... X_n$ are i.i.d, then we define **sample mean** $\bar{X} = \frac{1}{n}(X_1 + X_2 + ... X_n)$, if $\mathbb{E}(X) = \mu, \text{Var}(X) = \sigma^2$, then we can see $\mathbb{E}(\bar{X}) = n\mu, \text{Var}(\bar{X}) = \dfrac{\sigma^2}{n}$. Hence

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X} - \mu| > c) = 0$$

Using Chebyshev inequality, note that this means "everything will get close to the mean if we do a lot of experiment and take averages"

7 Random Variables and Conditional Expectation

Here we present some useful fact of random variables (denoted by X and Y).

Theorem 7.1. If X, Y are independent and $X \sim \text{Poi}(\lambda), Y \sim \text{Poi}(\mu)$, then $X + Y \sim \text{Poi}(\lambda + \mu)$

Theorem 7.2. If X, Y are independent and $X \sim \text{Gamma}(\alpha, \lambda), Y \sim \text{Gamma}(\beta, \lambda)$, then

$$X + Y \sim \text{Gamma}(\alpha + \beta, \lambda)$$

Theorem 7.3. If $X \sim U(0, 1)$, in order to generate Y using the formula $Y \sim g(X)$, then $g(x) = F_Y^{-1}(y)$.

Theorem 7.4. If $X \sim \mathcal{N}(\mu, \sigma^2)$, then the transformation $Z \sim \dfrac{X - \mu}{\sigma}$ will result in $Z \sim \mathcal{N}(0, 1)$

Theorem 7.5. Let $X \sim \mathcal{N}(\mu, \sigma^2)$, then for $\alpha, \beta \in \mathbb{R}$:

$$X + \alpha \sim \mathcal{N}(\mu + \alpha, \sigma^2) \qquad \text{and} \qquad \beta X \sim \mathcal{N}(\beta\mu, \beta^2\sigma^2)$$

Theorem 7.6. If the two random variables $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y \sim \mathcal{N}(\varphi, \upsilon^2)$ are independent, then $X + Y \sim \mathcal{N}(\mu + \varphi, \sigma^2 + \upsilon^2)$.

Definition 7.7. The **standard normal** random variable is $Z \sim \mathcal{N}(0, 1)$, have the distribution

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

we will denote the distribution function $F_Z(z)$ by $\Phi(z)$ ($\Phi(z)$ can be calculated in R using `pnorm(z)`)

Theorem 7.8 (Central Limit Theorem)**.** If $X_1, X_2 ... X_n$ are i.i.d random variables, then if $\mathbb{E}(X) = \mu, \text{Var}(X) = \sigma^2$

$$\lim_{n\to\infty} \mathbb{P}\left(a < \frac{X_1 + X_2 + ... + X_n - n\mu}{\sqrt{n\sigma^2}} < b\right) = \Phi(b) - \Phi(a)$$

Corollary 7.8.1 (DeMoivre–Laplace Theorem)**.** If $X \sim \text{Bio}(n, p)$, then

$$\lim_{n\to\infty} \mathbb{P}\left(a < \frac{X - np}{\sqrt{np(1-p)}} < b\right) = \Phi(b) - \Phi(a)$$

Note, the CLT essentially means doing any experiment X for a lot of times, the distribution will tends to $\mathcal{N}(\mu, \sigma^2)$, which is why the normal distribution is so important in statistics. DeMoivre Laplace theorem follows by the fact that doing multiple Bernoulli trials are Binomial.

Definition 7.9. Let $A(y)$ be a function of y defined by

$$A(y) = \mathbb{E}(X|Y = y) = \sum_x x \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

This function have a domain of values that Y can take (outputs of Y).

Definition 7.10. The **conditional expectation** $\mathbb{E}(X|Y)$ is $A(Y)$, note that this is a random variable.

Definition 7.11. The **conditional variance** of X given Y is

$$\text{Var}(X|Y) = \mathbb{E}(X^2|Y) - [\mathbb{E}(X|Y)]^2$$

It is also a random variable.

Theorem 7.12 (Tower Law)**.** For any random variable $X, Y, \mathbb{E}(X) = \mathbb{E}[\mathbb{E}(X|Y)]$.

Theorem 7.13 (Total Variance)**.** For any random variable $X, Y, \text{Var}(X) = \text{Var}[\mathbb{E}(X|Y)] + \mathbb{E}[\text{Var}(X|Y)]$.

8 Moment Generating Function

Definition 8.1. The rth **moment** of a random variable X is $\mathbb{E}(X^r)$

Definition 8.2. The **moment generating function (MGF)** of a random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}] = \begin{cases} \sum_x e^{tx} p_X(x) & \text{when } X \text{ discrete} \\ \int_{-\infty}^{\infty} e^{tx} f_X(x) dx & \text{when } X \text{ continuous} \end{cases}$$

Theorem 8.3. Since

$$M_X(t) = 1 + t\mathbb{E}(X) + \frac{t^2}{2}\mathbb{E}(X^2) + \frac{t^3}{3!}\mathbb{E}(X^3) + ...$$

note that $M_X(0) = 1, M_X'(0) = \mathbb{E}(X), M_X''(0) = \mathbb{E}(X^2)$, the rth derivative at $t = 0$ recovers the rth moment. That is, $M_X^{(r)}(0) = \mathbb{E}(X^r)$. This is especially useful for finding the variance given a MGF.

Theorem 8.4. If $X_1, X_2 ... X_n$ are independent, then if $Z = X_1 + X_2 + ... + X_n$

$$M_Z(t) = \prod_{i=1}^{n} M_{X_i}(t)$$

Definition 8.5. A **random sum** is $S = X_1 + X_2 + ... + X_N$, where $X_1, X_2 ... X_N$ are i.i.d and N is a discrete random variable who takes non-negative integer values. For example X can be the amount of payout in a game and N can be the number of winning players in a game.

Theorem 8.6. In the above setting, we have $\mathbb{E}(S) = \mathbb{E}(N)\mathbb{E}(X)$ and $M_S(t) = M_N[\log(M_X(t))]$, where $\log(x)$ is the natural logarithm base e.

MGF are extremely useful due to the following theorem.

Theorem 8.7 (Uniqueness of MGF)**.** If for an interval containing the origin $I \subseteq \mathbb{R}$, and for all $t \in I$, we have

$$M_X(t) = M_Y(t)$$

for two random variables X and Y, then $F_X(x) = F_Y(x)$. Equivalently, they have the same density.

Here are some MGF of famous distribution. Note that the function will be undefined in some place due to the constraint $e^x > 0$:

- $X \sim \text{Bernoulli}(p) : M_X(t) = (1-p) + pe^t$
- $X \sim \text{Bio}(n, p) : M_X(t) = [(1-p) + pe^t]^n$
- $X \sim \text{Geom}(p) : M_X(t) = \dfrac{pe^t}{1-(1-p)e^t}$ for $(1-p)e^t < 1$
- $X \sim \text{Poi}(\lambda) : M_X(t) = e^{\lambda(e^t - 1)}$
- $X \sim U(\alpha, \beta) : M_X(t) = \dfrac{e^{\beta t} - e^{\alpha t}}{t(\beta - \alpha)}$
- $X \sim \text{Exp}(\lambda) : M_X(t) = \dfrac{\lambda}{\lambda - t}$ for $\lambda > t$
- $X \sim \text{Gamma}(\alpha, \lambda) : M_X(t) = \left(\dfrac{\lambda}{\lambda - t}\right)^{\alpha}$ for $\lambda > t$
- $X \sim \mathcal{N}(\mu, \sigma^2) : M_X(t) = \exp\left(t\mu + \dfrac{t^2}{2}\sigma^2\right)$